



Instacart Case Study: Feedback Loops in Grocery Recommender Systems

Kevin O'Neill & Cameron Wang

11 May 2026

Physical vs Online Grocery Shopping

What are some differences between the physical and online grocery shopping experiences?



←

Sort ▾ Dietary ▾ Ingredient Preferences ▾

Results for "butter"



\$12.99
Kerrygold Salted Butter Sticks
16 oz - 2 sizes
Many in stock
Sponsored



\$4.79
Store Brand Butter, Sweet Cream, Salted
4 ct
Many in stock



\$12.99
Kerrygold Pure Irish Butter
14 oz
Many in stock
Sponsored



\$12.99
Kerrygold Unsalted Pure Irish Butter
4 ct - 2 sizes
★★★★★ (6)
Many in stock

Grass fed

Physical vs Online Grocery Shopping

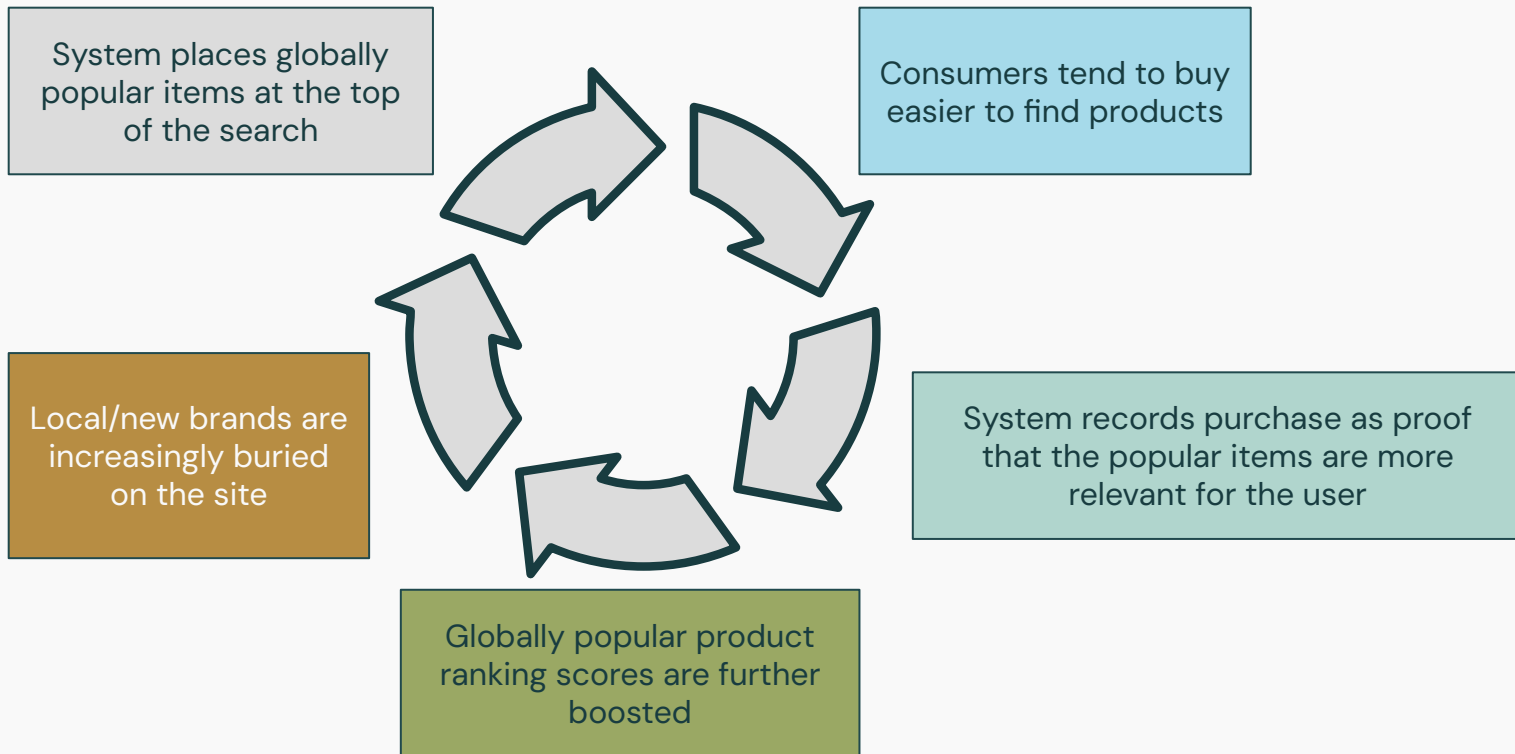
Physical

- Users discover products through sensory cues, unplanned discovery
- Strategic placement decisions are made by the grocery store and/or brands themselves
- Shelf space is fixed, so there exists a baseline level of variety
- Forced exposure to unrelated categories
- Relatively more difficult for customers to compare prices between products
- **Everyone sees the same products in the same way (to an extent)**

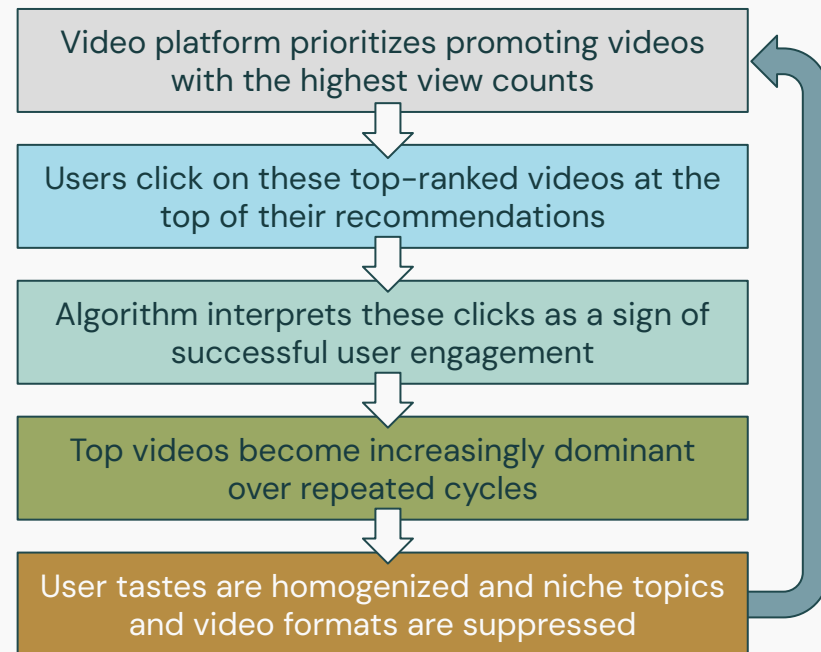
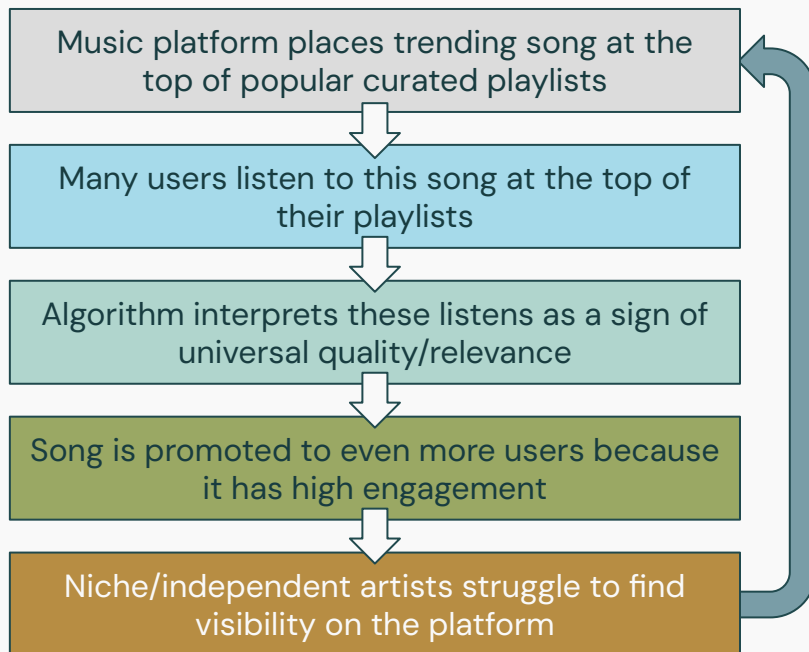
Online

- Product discovery is primarily search-driven
- Digital shelf space is infinite but filtered (only see a few products on the page)
- Experience is optimized for efficiency (ex: Buy it Again, Past Purchases)
- Easier to dynamically price items and **personalize recommendations**

Feedback Loops for Recommender Systems



Feedback Loops for Recommender Systems



Dataset and Methods

Dataset: Instacart prior-order dataset (200K+ anonymized users across 3.2M+ user-orders)

Observations Contain

- User ID
- Order Number (Chronological)
- List of Products Purchased
 - Product ID, Aisle, Department
- Global Purchase Count (across dataset)

Popularity Tiers

- popular → top 10% by global purchase count
- ultra-popular → top 1% by global purchase count

Outcome Metrics

- Popular-item share $P(t)$: fraction of items in the basket belonging to the popular tier.
- Ultra-popular-item share $U(t)$: fraction belonging to the ultra-popular tier.
- Mean log global popularity $L(t)$: average of $\log(\text{global purchase count})$ across all items in the basket.
- Within-basket HHI $H(t)$: Herfindahl-Hirschman Index over product-type shares within the basket, measuring internal basket concentration (lower = more diverse).

Key Findings

+3.4%

relative increase in popular item share from
orders 1 to 20

+8.8%

relative increase in ultra-popular item share from
orders 1 to 20

-4.0%

drop in internal basket diversity (HHI)

±26%

spread of popular-item share change between
top and bottom 10% of users

Main Takeaway: Algorithmic “rich-get-richer” feedback loop causes decrease in basket diversity and a shift towards popular and ultra-popular items over time

Who is affected by this?

Consumers

- Recommendation systems shape what users actually see and buy
- Users may gradually be pushed toward the same mainstream products
- Niche dietary, cultural, or specialty preferences become harder to discover
- Over time, users may lose product diversity without realizing it

Producers

- Large established brands benefit because they already have strong purchase history
- Small/local/niche producers struggle to gain visibility
- New products may never get enough exposure to compete fairly
- Producers must pay for visibility

Why is this important?

The real impact on you as an Instacart user



Your Feed is Curated, Not Neutral

Every search result and "recommended" product is shaped by your past purchases + paid placements. You never see a full, unbiased shelf.



You Pay More Without Knowing

Sponsored products dominate top results regardless of price or quality. The algorithm optimizes for revenue, not your best deal.



Choice Diversity Shrinks Over Time

The feedback loop narrows what you see with every order, locking you into a smaller product world and reducing your awareness of alternatives.

Instacart's recommendation system shapes your diet, spending and health choices.

Recommendations

For Shoppers

Search by ingredient or keyword

Bypass recommendations entirely by typing exactly what you want in the search bar.

Sort by "Price" or "Rating" NOT "Relevance"

Reordering results breaks the algorithm's sponsored-first ranking.

Explore "New" or "Local" filters

Actively seek new brands to break the feedback loop reinforcing old habits.

For Instacart

Diversity Score in Recommendations

Introduce algorithmic diversity quotas like ensure at least 20–30% of recommended items come from non-sponsored or emerging brands.

Transparent Labeling

Clearly mark all sponsored/promoted products so users understand what is paid placement vs. organic results.

"Explore Mode" Toggle

Offer users an opt-in mode that intentionally surfaces products outside their purchase history.

Bibliography

Celma, Ò. (2010). *Music recommendation and discovery: The long tail, long fail, and long play in the digital music space*. Springer.

Instacart. (2017). The Instacart Online Grocery Shopping Dataset 2017. Kaggle.

<https://www.kaggle.com/datasets/psparks/instacart-market-basket-analysis>

Mansoury, M., Abdollahpouri, H., Pechenizkiy, M., Mobasher, B., & Burke, R. (2020). Feedback loop and bias amplification in recommender systems. *Proceedings of the 29th ACM International Conference on Information and Knowledge Management*, 2145–2148. <https://doi.org/10.1145/3340531.3412063>